

$SU(2)_L$ Chirality from the Galois Structure:

Su/Sd Separation and $Cl(6)$ Grade Parity

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Status markers

- [K]** *Confirmed* — algebraically complete.
- [B]** *Established* — secured by script.
- [S]** *Speculative* — conjecture, still open.

Abstract

We examine whether the $SU(2)_L$ chirality structure of the Standard Model follows from the $Cl(6)$ grade parity established in Doc. 341. The pseudoscalar $\gamma_5 = e_1 e_2 e_3 e_4 e_5 e_6$ acts on grade r as the grade involution, $\gamma_5 X \gamma_5^{-1} = (-1)^r X$; this is not an eigenvalue equation on blades, and in $Cl(6, 0)$, $\gamma_5^2 = -1$. Chirality has to be captured by a projector $P_{\pm} = (1 \pm i\gamma_5)/2$ in the complexification. The Su/Sd split (Doc. 344) is therefore not directly the right/left split: by Doc. 344, Su contains even and odd N values, Sd[0] is the anti-neutrino, and $Cl(6)$ has no grade 7 for e_L . The $SU(2)_L$ chirality structure left open in Docs. 347/348 remains open **[S]**.

Background

Doc. 344 **[B]**: the jE7 projector separates the Su and Sd sectors, with $N(\text{Su}) = [0, 1, 1, 1, 2, 2, 2, 3]$ (even and odd values) and negative $N(\text{Sd})$; Su[1–3] = anti-down, Su[4–6] = up, Su[7] = positron, Sd[0] = anti-neutrino, Sd[7] = electron. A split “Su: N even, Sd: N odd” does not exist; indexing and N values below follow Doc. 344.

1 Theorem A — Grade parity and γ_5 **[K]**

$\gamma_5 X \gamma_5^{-1} = (-1)^r X$ for a grade- r element $X \in Cl(6)$, since γ_5 anticommutes with every generator e_i , so that $\gamma_5 X = (-1)^r X \gamma_5$ for a grade- r blade. **[K]** This is a conjugation statement (grade involution), not an eigenvalue equation on blades: left multiplication by γ_5 maps grade r to grade $6 - r$ (e.g. $\gamma_5 \cdot 1 = \gamma_5 \neq 1$). Moreover, in $Cl(6, 0)$ with $e_i^2 = +1$: $\gamma_5^2 = (-1)^{15} = -1$. Chirality has

to be captured via conjugation or a projector, e.g. $P_{\pm} = (1 \pm i\gamma_5)/2$ in the complexification since $(i\gamma_5)^2 = +1$. An assignment "even grades = right-handed, odd grades = left-handed" as a γ_5 eigenvalue on blades therefore does not hold.

2 Theorem B — Su sector and Cl(6) grades

"Su = even grades = right-handed" does not hold: by Doc. 344 the Su sector contains even and odd N values, and even read by N classes, $|N| = 3$ in Su is the positron (not e_R). In addition, by Theorem A the grade carries no γ_5 eigenvalue.

3 Theorem C — Sd sector and Cl(6) grades

Likewise "Sd = odd grades = left-handed" does not hold: by Doc. 344, Sd[0] is the anti-neutrino and Sd[7] the electron, and read by N classes, $|N| = 2$ in Sd is anti-up (not $\bar{d}$). Cl(6) has only grades 0 to 6 with 1, 6, 15, 20, 15, 6, 1 blades, 2^6 in total; there is no grade 7, not even as a multigrade. An assignment grade = $2N + 1$ therefore breaks down at $N = 3$; the assignment of e_L is open **[S]**.

4 Theorem D — SU(2)_L doublets (open)

A doublet assignment $(\nu_L, e_L) = (\text{Sd}[0], \text{Sd}[3])$ does not hold, because Sd[0] is the anti-neutrino by Doc. 344 and the assignment of e_L is open (Theorem C). Whether SU(2)_L acts on exactly one of the two sectors, and which states form left-handed doublets and right-handed singlets, remains open **[S]**; the question has to be posed with the projector $P_{\pm}$ (Theorem A) and with the indexing and N values of Doc. 344.

5 Theorem E — $\mathbb{Z}_3$ fixed points/orbit and grade parity

From Doc. 341: the σ rotation separates Vss[0, 2, 4] (σ -fixed points, even grades) from Vss[1, 3, 5] (σ -three-orbit, odd grades) **[B]**. Identifying these classes with right- and left-handed presupposes the eigenvalue reading of the grade parity and, by Theorem A, does not hold in this form. Whether the $\mathbb{Z}_3$ symmetry of $T^4/\mathbb{Z}_3$ forces the chirality split via $P_{\pm}$ is open **[S]**.

6 Summary

Theorem	Content	Status
A	$\gamma_5 X \gamma_5^{-1} = (-1)^r X$ (grade involution, no eigenvalue)	[K]
B	Su sector $\neq$ even grades (Doc. 344)	—
C	Sd sector $\neq$ odd grades; no grade 7, e_L open	[S]
D	SU(2) _L doublets from the Su/Sd sector	[S]
E	$\mathbb{Z}_3$ fixed points/orbit = even/odd grades; chirality open	[S]

The grade parity of Doc. 341 is captured via γ_5 as the grade involution **[K]**; the SU(2)_L chirality structure is not thereby derived. Open **[S]**: the chirality assignment via the projector $P_{\pm}$ with

indexing and N values from Doc. 344, the assignment of e_L , the $SU(2)_L$ doublet structure, and the exact CKM mixing angles and CP phase δ_{CP} (Doc. 348).

Verification script

`pruef_349_chiralitaet.py` — checks the signs $(-1)^r$ of the grade parity; it does not support the chirality and sector assignments of Theorems B–E.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations: Johann Pascher (ORCID 0009-0000-6518-4064).

Bibliography

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